# CSEP 590tv: Quantum Computing

**Dave Bacon**

Aug 10, 2005

**Today's Menu**

*Entanglement*

*Bell's Theorem*

*Entanglement and Quantum Computation*

*Quantum Communication Complexity*

*Entanglement as a resource*

# Administrivia

Questions on final?

Last lecture will be a non-technical summary of the sections of quantum information science we didn't cover.

Quantum cryptography, quantum error correction, quantum communication, physical implementations.

Plus probably a little something on some far off topics like quantum information and the black hole information paradox.

# Email From Home

From: Nancy Bacon
To: Dave Bacon
Subject: teaching
Parts/Attachments:
  1   OK     1 lines  Text
  2 Shown    0 lines  Text
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from you earthly mortal mother,  i learn best when someone gives me a broad and general concept, then goes into details, and then returns to the basic concept, of course the trick is to find just the very right details to explain the general concept. ha!  good luck davers

# Big Picture

Entanglement has long been known to be one of the
fundamentally strange things about quantum theory.

For years, people worried about entanglement.
(Einstein, Schrodinger, Bell,….)

What happening in quantum computing is that people stopped learning to worry about entanglement, and began to realize that if they just accepted it, it was a very valuable
resource!

Accept Entanglement!

But what is entanglement? Why is it “mysterious”?
Why is it important for quantum computation?

# Bipartite Entanglement

$|v\rangle$



Alice's qubit

Bob's qubit



Two qubits have a wave function $|v\rangle$ which is either

Separable: we can express it as

$$|v\rangle = |a\rangle \otimes |b\rangle$$

$|a\rangle, |b\rangle$ valid wave functions

Entangled: we cannot express it as $|v\rangle = |a\rangle \otimes |b\rangle$

$$|v\rangle \neq |a\rangle \otimes |b\rangle$$

# Two Qubits, Separable

Example : $|v\rangle = |a\rangle \otimes |b\rangle$

$$|a\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle \quad |b\rangle = \frac{i}{\sqrt{5}}|0\rangle + \frac{2}{\sqrt{5}}|1\rangle$$

$$|v\rangle = |a\rangle \otimes |b\rangle = \left(\frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle\right) \otimes \left(\frac{i}{\sqrt{5}}|0\rangle + \frac{2}{\sqrt{5}}|1\rangle\right)$$

$$= \frac{1}{2}\frac{i}{\sqrt{5}}|0\rangle \otimes |0\rangle + \frac{1}{2}\frac{2}{\sqrt{5}}|0\rangle \otimes |1\rangle + \frac{\sqrt{3}}{2}\frac{i}{\sqrt{5}}|1\rangle \otimes |0\rangle + \frac{\sqrt{3}}{2}\frac{2}{\sqrt{5}}|1\rangle \otimes |1\rangle$$

$$= \frac{i}{2\sqrt{5}}|00\rangle + \frac{1}{\sqrt{5}}|01\rangle + \frac{\sqrt{3}i}{2\sqrt{5}}|10\rangle + \frac{\sqrt{3}}{\sqrt{5}}|11\rangle$$

# Two Qubits, Entangled

Example:

$$|v\rangle = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{\sqrt{3}}{2} \end{bmatrix} = \frac{1}{2}|00\rangle + \frac{\sqrt{3}}{2}|11\rangle$$

Assume

$|v\rangle = |a\rangle \otimes |b\rangle$ $\quad |a\rangle = a_0|0\rangle + a_1|1\rangle \quad |b\rangle = b_0|0\rangle + b_1|1\rangle$

$$= a_0b_0|00\rangle + a_0b_1|01\rangle + a_1b_0|10\rangle + a_1b_1|11\rangle$$

$a_0b_1 = 0$

Either

$a_0 = 0$ but this implies $a_0b_0 = 0$

or

$b_1 = 0$ but this implies $a_1b_1 = 0$

contradictions

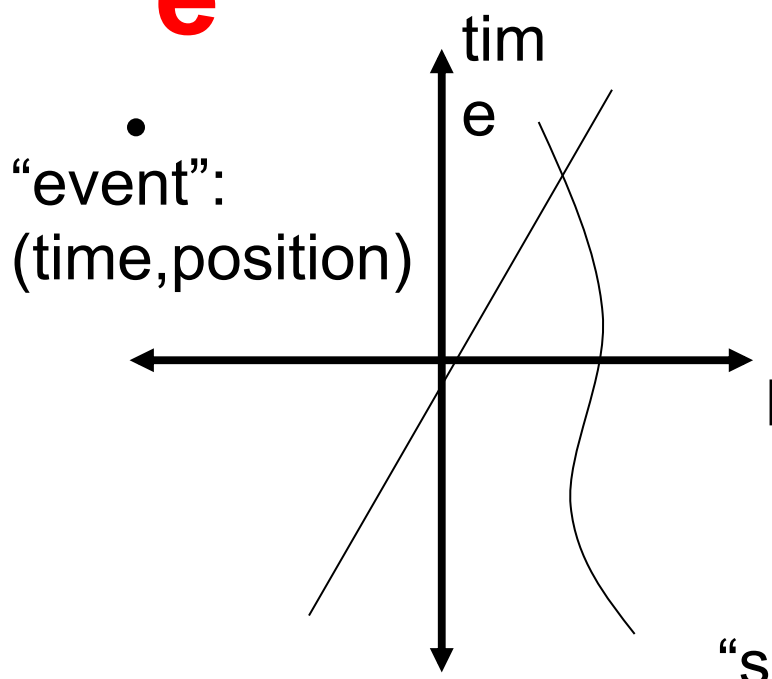
So $|v\rangle$ is not a separable state. It is entangled

# Special Relativity

To understand what makes entanglement so interesting in physics we need to know a little special relativity.

We don’t need to know how to calculate in special relativity, but we do need to understand the concept of “locality” which
arises in special relativity.

# Spacetime




“event”:
(time,position)

“inertial frame”:
constant velocity

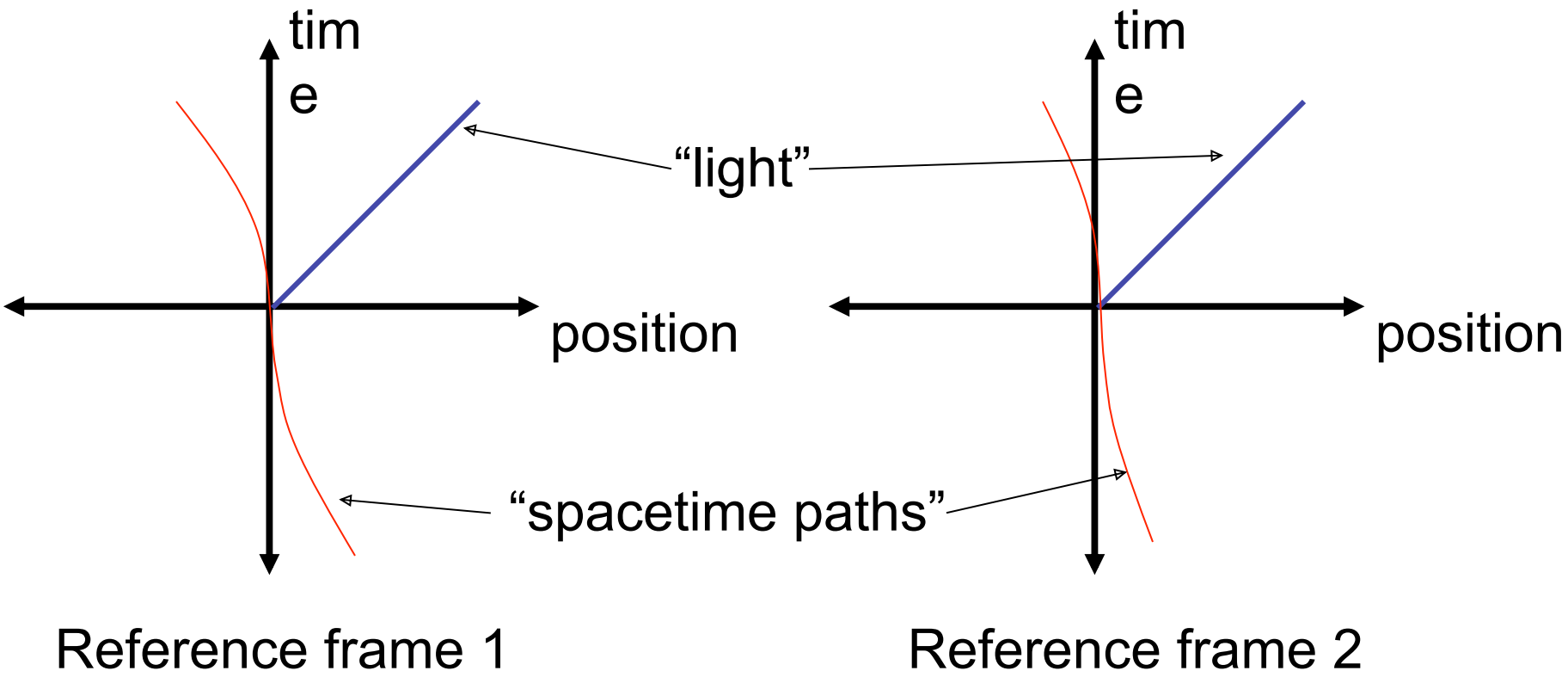
“spacetime path”:
curve in
spacetime

# Special Relativity

Special relativity:

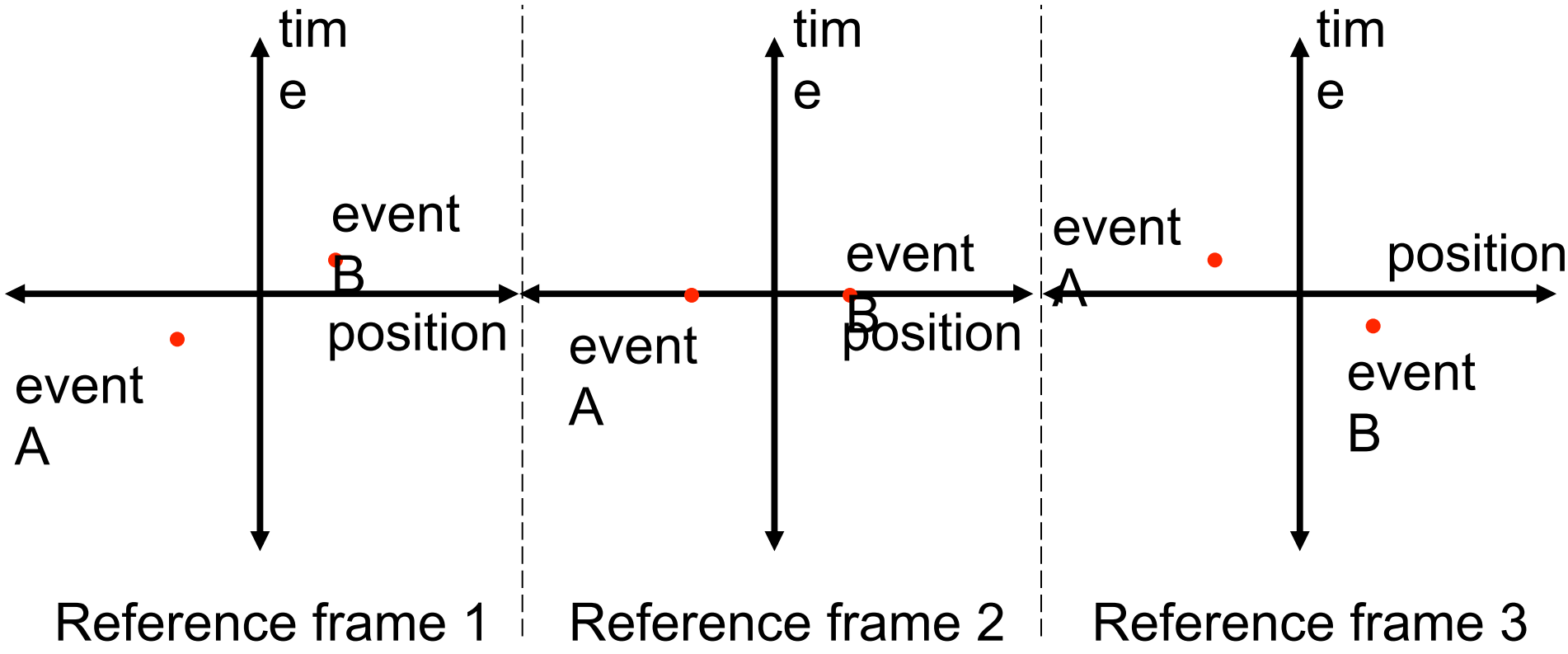
(1) physics is the same in all inertial reference frames

(2) speed of light is same in all reference frames.



# Simultaneity

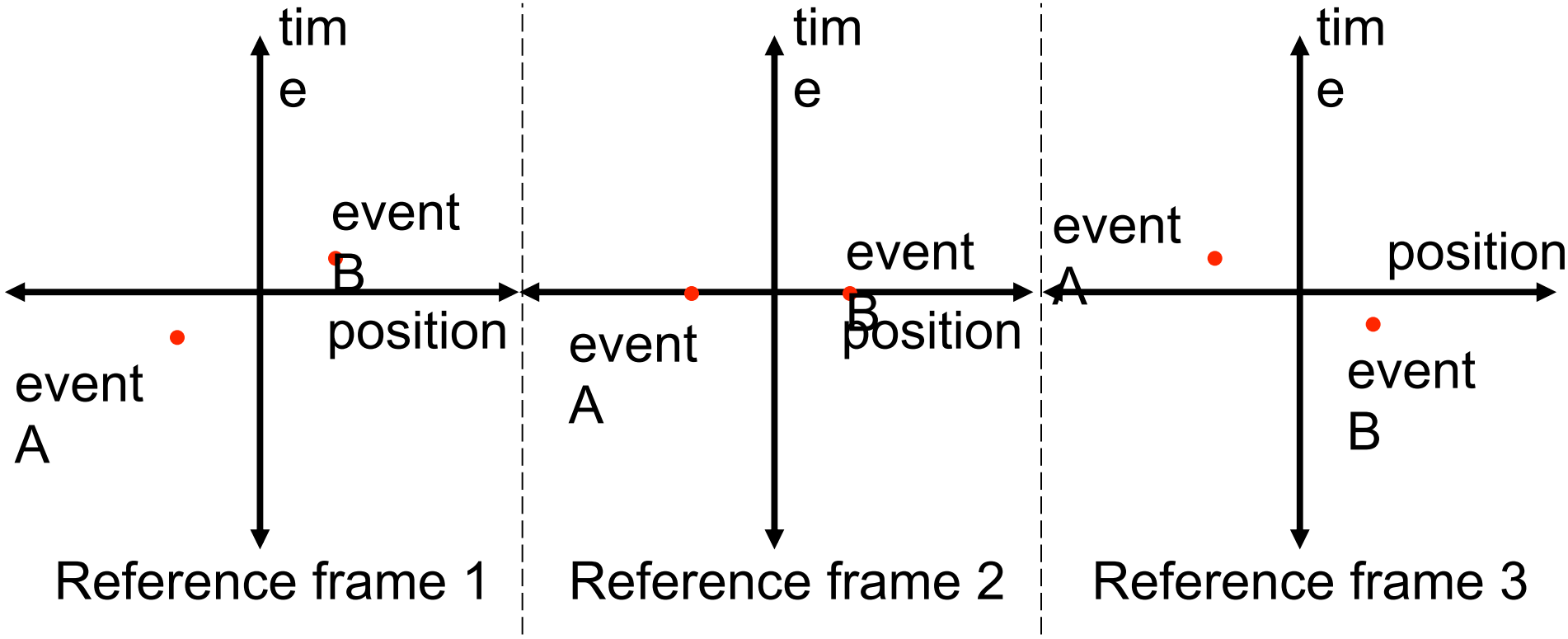
In special relativity, the idea of simultaneity is relative:




Different events are seen as occurring in a different order, depending on the reference frame.

# Special Relativity

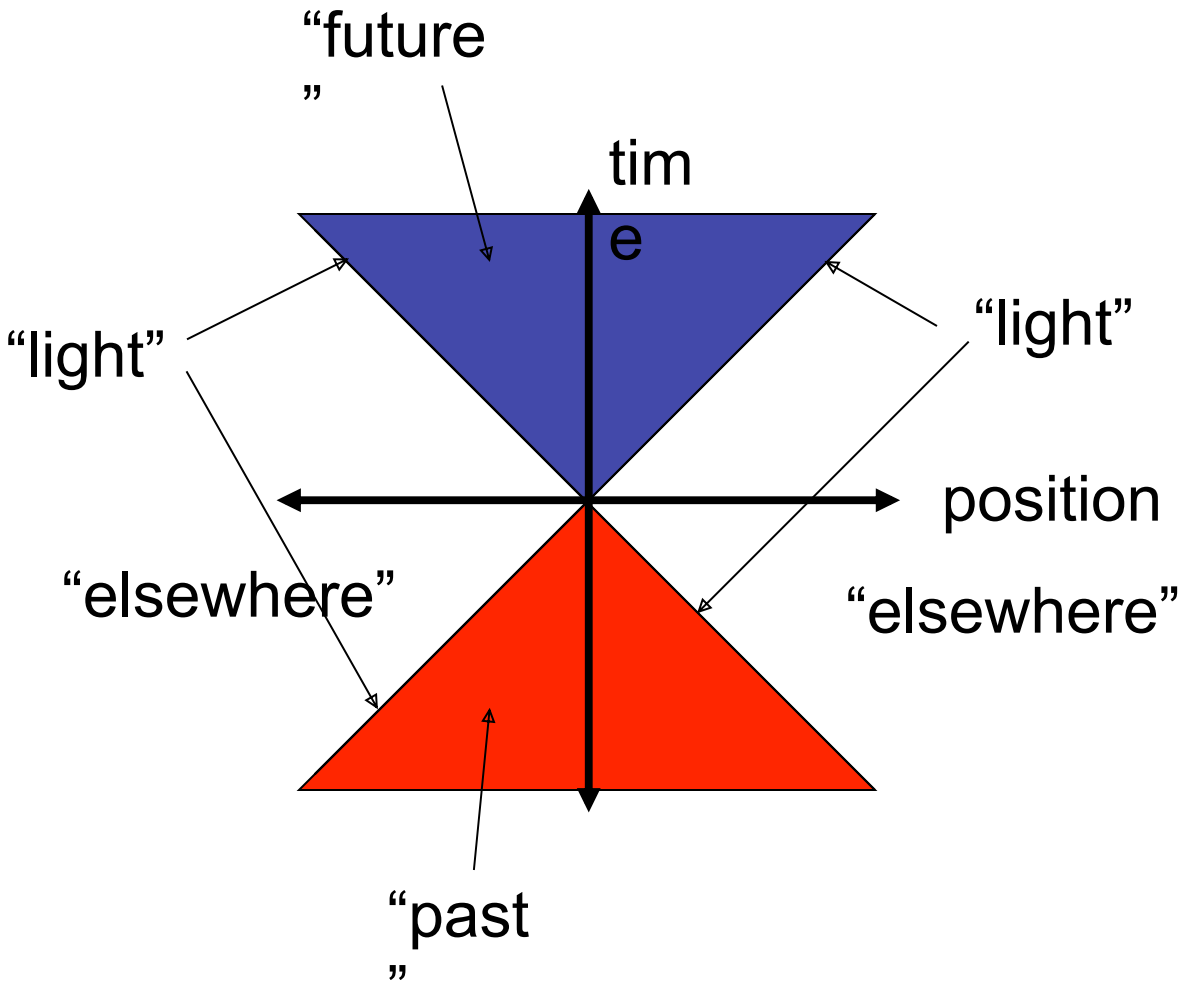
Signal: I do something at my location and time such that you can, conditional on what I do, act conditionally at your location and time.




If I try to send a signal from A to B, then in some reference frame, B acts before A sends the signal. Better not allow this!

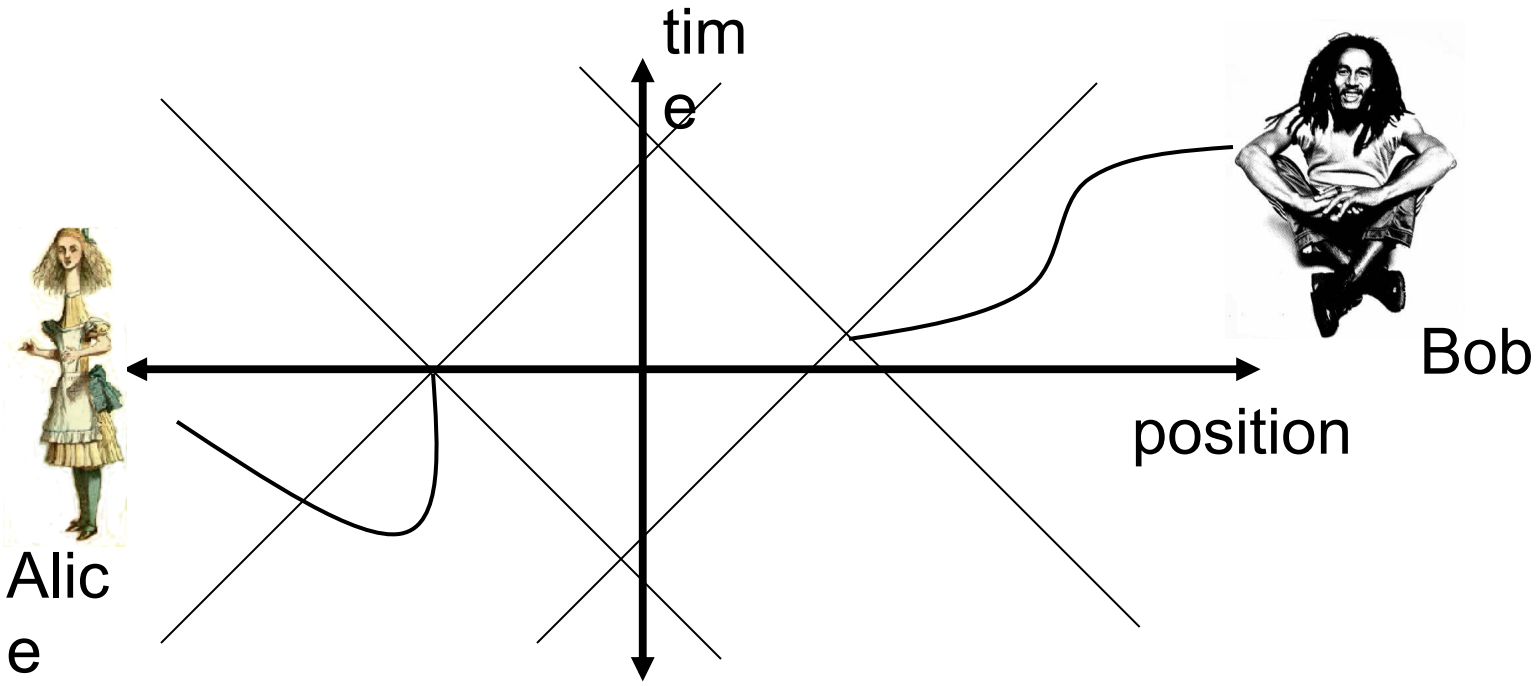
# Special Relativity

For all inertial observers moving through the origin, there are three regions of spacetime which are preserved between inertial frames:



# Special Relativity

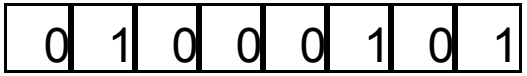
When Alice and Bob both live in each other's "elsewhere" they cannot communicate with each other:



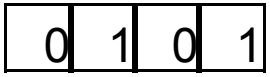

We call such setups "space-like separated"

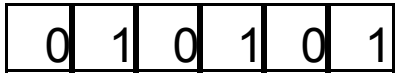
# Cellular Automata

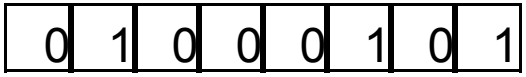
Cellular automata (CA):

State: 

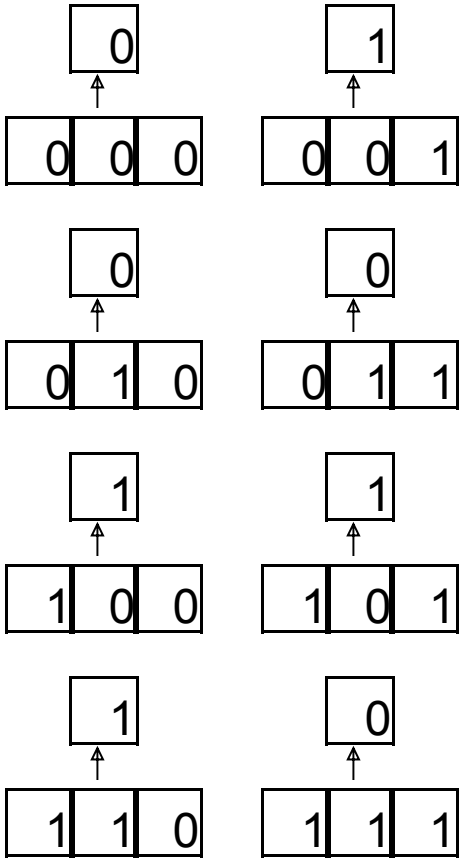

Evolution of a CA:

T= 2 


T= 1 


T= 0 

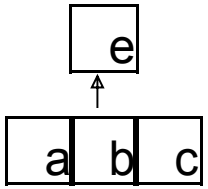

CA Rules

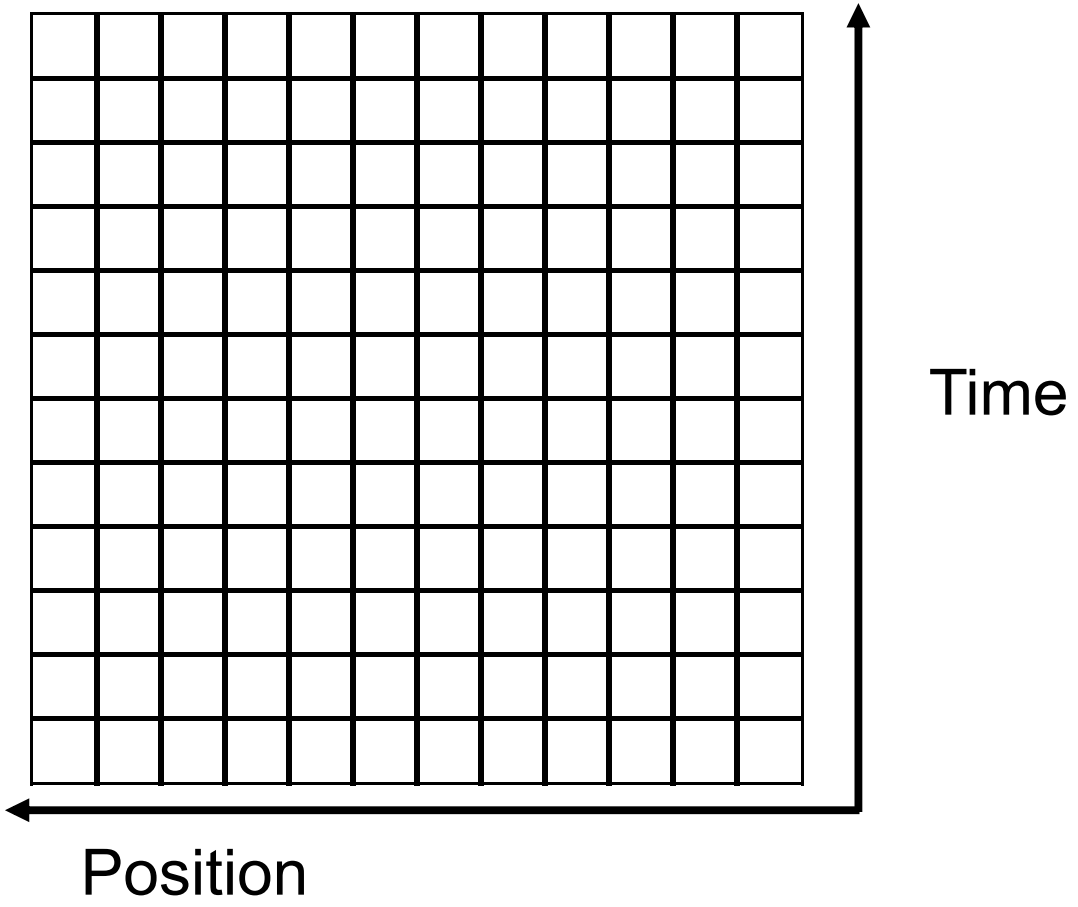



These rules are <u>local</u>

# Cellular Automata

CA rules are local

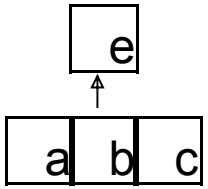



CA “Spacetime”

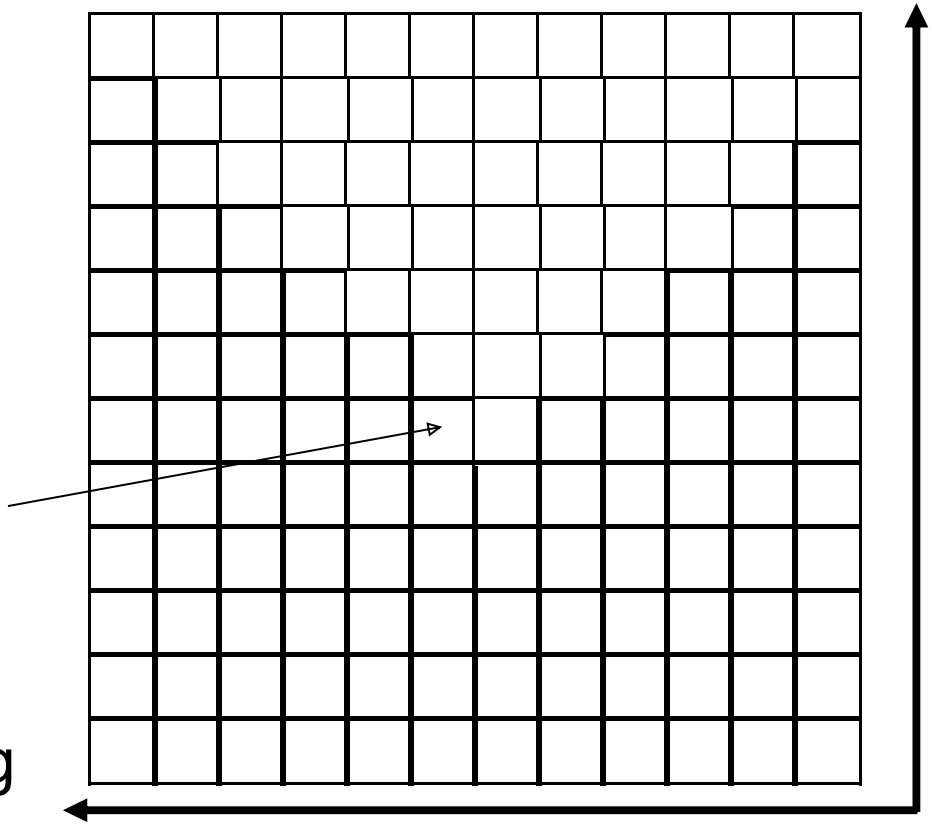
# Cellular Automata

CA rules are local




Changing this value can only every change
the blue states (the future)

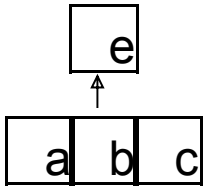
Finite speed of signaling




CA “Spacetime”

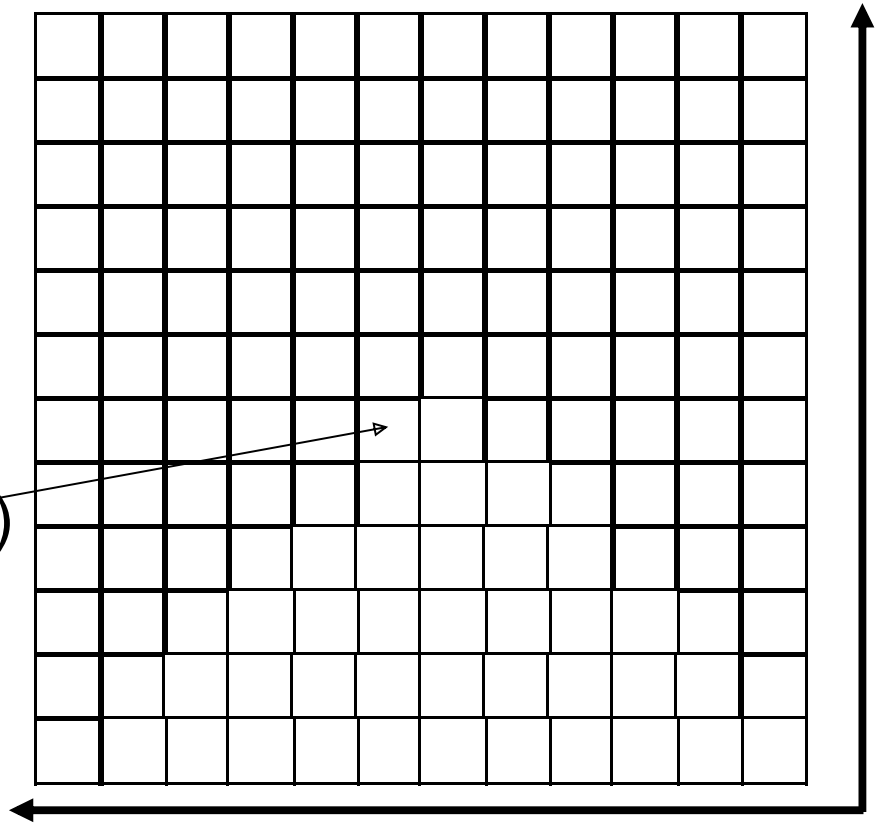
# Cellular Automata

CA rules are local




This state is a function of the states in red (the past)

Finite speed of signaling




CA “Spacetime”

# Bipartite Entanglement





Alice's qubit



Bob's qubit



We will be interested in situations where Alice and Bob are spacelike separated. It is in these setups, where they cannot communicate, that quantum theory becomes interesting.

# Separable States: Properties



Alice's qubit

Bob's qubit



If Alice acts on her qubit, while Bob does nothing, the resulting unitary is of the form

$$U \otimes I$$

If Bob acts on his qubit, while Alice does nothing, the resulting unitary is of the form

$$I \otimes V$$

Both of these preserve se

$$(U \otimes I)|a\rangle \otimes |b\rangle = (U|a\rangle) \otimes (I|b\rangle) = |a'\rangle \otimes |b\rangle$$

What about entangled states?

$$|a'\rangle = U|a\rangle$$

# Schmidt Decomposition



Alic
e

$d_a$

$d_b$

Bob



All bipartite quantum systems can be expressed as

$$|v\rangle = \sum_{i=0}^{\min(d_A,d_b)} \sqrt{p_i}|i\rangle_A \otimes |i\rangle_B \qquad \sum_{i=0}^{\min(d_A,d_b)} p_i = 1$$

For some choice of basis of the two parties, $|i\rangle_A$ and $|i\rangle_B$

Example:

$$|v\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle$$

$$|v\rangle = \frac{1}{2}(|0\rangle - |1\rangle) \otimes |1\rangle + \frac{1}{2}(|0\rangle + |1\rangle) \otimes |0\rangle$$

$$|v\rangle = \frac{1}{\sqrt{2}}|-\rangle \otimes |1\rangle + \frac{1}{\sqrt{2}}|+\rangle \otimes |0\rangle \quad |\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$$

# Schmidt Decomposition



Alic
e

$d_a$

$d_b$

Bob



All bipartite quantum systems can be expressed as

$$|v\rangle = \sum_{i=0}^{\min(d_A,d_b)} \sqrt{p_i}|i\rangle_A \otimes |i\rangle_B \qquad \sum_{i=0}^{\min(d_A,d_b)} p_i = 1$$

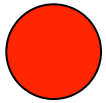
For some choice of basis of the two parties, $|i\rangle_A$ and $|i\rangle_B$

When only one $p_i$ is nonzero (and hence equal to one) then
the wave function is separable

Thus we can use Schmidt form to test for entanglement.

# Entangled States: Properties





Alice's qubit



Bob's qubit



If Alice acts on her qubit, while Bob does nothing, the resulting unitary is of the form

$$U \otimes I$$

If Bob acts on his qubit, while Alice does nothing, the resulting unitary is of the form

$$I \otimes V$$

Both of these preserve entanglement.

# Entangled States: Properties



Alice's qubit

Bob's qubit



$$|v\rangle = \sum_{i=0}^{\min(d_A,d_b)} \sqrt{p_i}|i\rangle_A \otimes |i\rangle_B$$

$$(U \otimes I)|v\rangle = \sum_{i=0}^{\min(d_A,d_b)} \sqrt{p_i}(U|i\rangle_A) \otimes |i\rangle_B$$

Just a change of basis.
Does not change $p_i$

Hence $U \otimes I$ and $I \otimes V$ preserve entanglement

# Local Unitaries



Alice's qubit

Bob's qubit



Local unitaries

A $U$

B $V$

separable $U \otimes V$ separable

entangled $U \otimes V$ entangled

separable $U \otimes V$ entangled

# Entanglement Generation

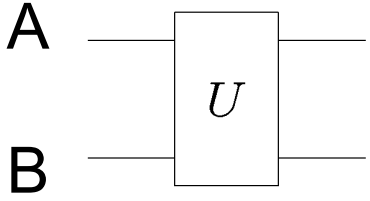


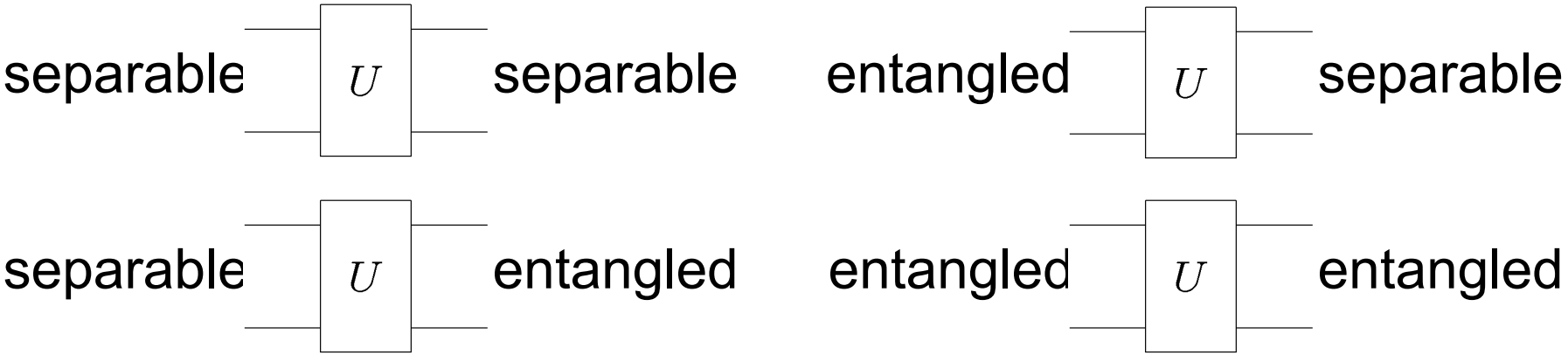
Alice's qubit

Bob's qubit



Two party unitary:

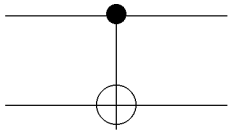


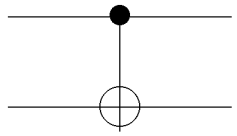



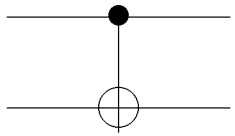

Depends on the details of the unitary and on the inputs

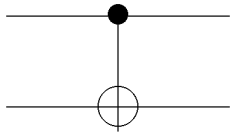
# Entanglement Generation

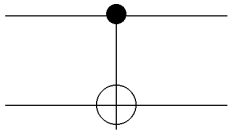
Example :



separable $|1\rangle$, $|0\rangle$  $|1\rangle$, $|1\rangle$ separable

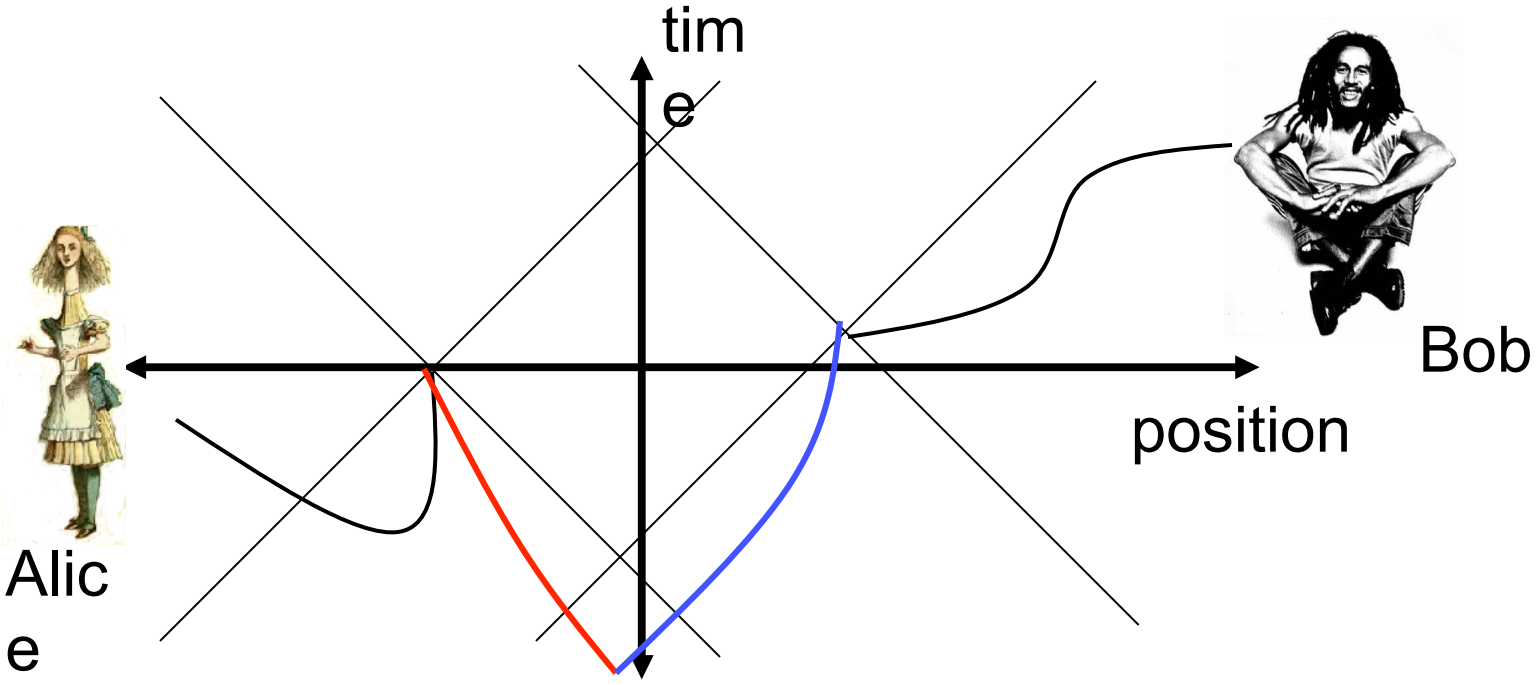
separable $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|0\rangle$  $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ entangled

entangled $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$  $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|0\rangle$ separable

entangled $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$  $\frac{1}{2}(|00\rangle + |01\rangle - |10\rangle + |11\rangle)$ entangled

$\frac{1}{\sqrt{2}}(|0\rangle \otimes |+\rangle + |1\rangle \otimes |-\rangle)$ $\frac{1}{\sqrt{2}}(|0\rangle \otimes |+\rangle - |1\rangle \otimes |-\rangle)$

# Entanglement Generation




Entanglement needs to be generated “locally”
i.e. the two parties got together in the past and interacted their qubits

# Correlation

$|v\rangle$





$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$



Alice's qubit

Bob's qubit



1. Alice measures in the computational basis.

Facts:

With probability 50% she gets outcome 0.
With probability 50% she gets outcome 1.

If her outcome was 0, the new wave function is $|00\rangle$
If her outcome was 1, the new wave function is $|11\rangle$

2. Bob now measures in the computational basis

If Alice's outcome was 0, and Bob's outcome will be 0
If Alice's outcome was 1, and Bob's outcome will be 1

# Correlation

$$|v\rangle$$



$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

Alice's qubit

Bob's qubit



If Alice's outcome was 0, and Bob's outcome will be 0
If Alice's outcome was 1, and Bob's outcome will be 1

Notice that this is NOT signaling. Why?

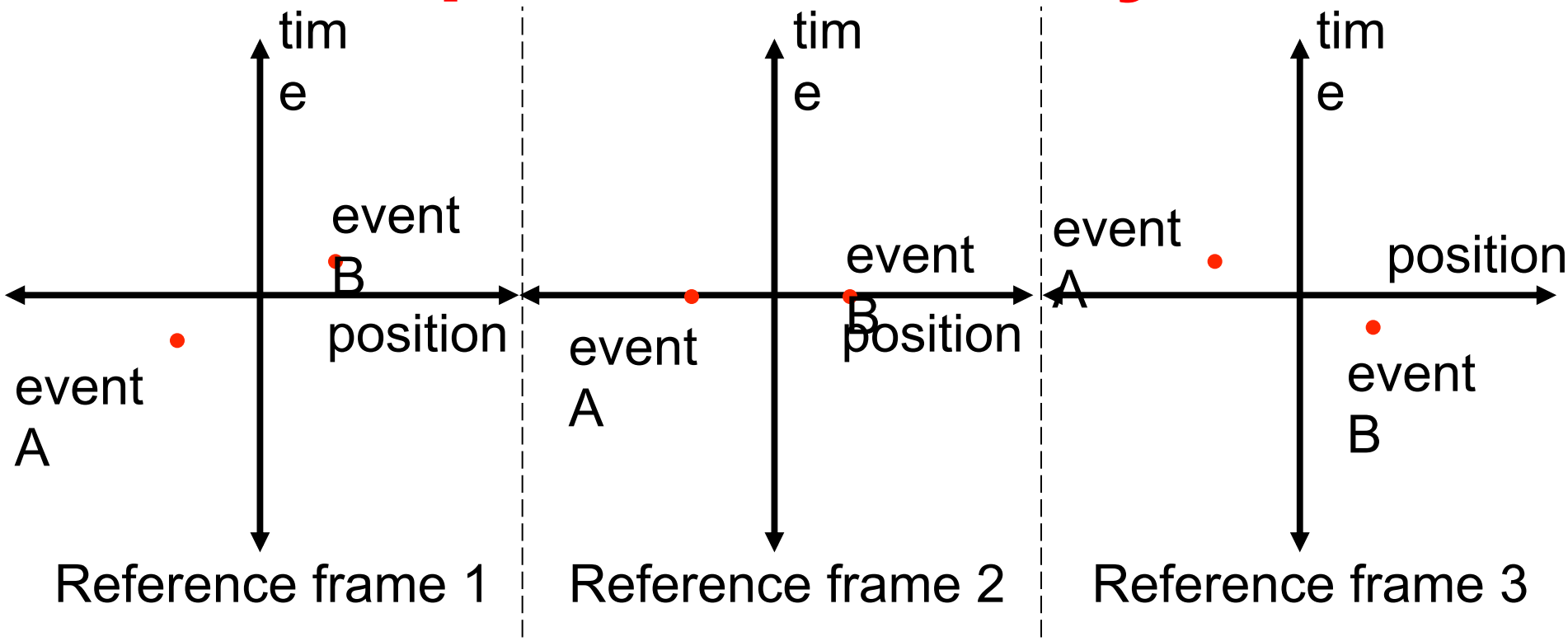
What does Bob see?

Having NOT seen Alice's measurement outcome, he finds that he gets

0 with 50% probability and

1 with 50% probability.

# Special Relativity




Should we worry that "who does the measurement first" depends on the frame of reference?

From the perspective of each party, NO, because they always get the same probabilities of outcomes.

# Correlation

$$|v\rangle$$





$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$





Alice's qubit

Bob's qubit

A completely classical way to simulate this experiment:

1. Flip a fair coin.
   If the result is heads, put 0 into two boxes.
   If the result is tails, put 1 into two boxes.
2. Give Alice and Bob the boxes.
3. Parties perform measurements by opening their boxes and reporting the classical bit inside as their measurement outcome

# Correlation

$$|v\rangle$$





$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$





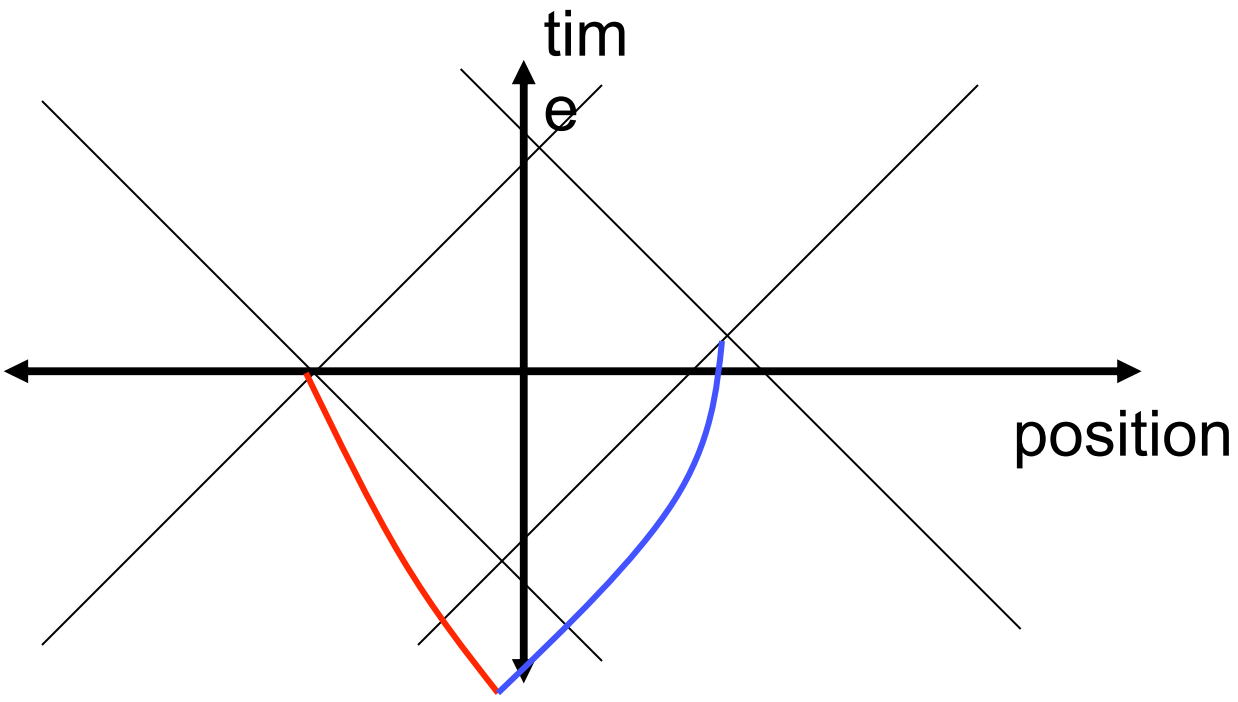
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# Local Hidden Variable Model




Suppose we try to respective special relativity, but at the same time, reproduce the probabilities we get from quantum systems

Local Hidden Variable Models

# Einstein, Bohr (1927-36)

After the creation of quantum theory, Einstein became very concerned about quantum theory.



Einstein: “God does not play dice.”
Bohr: “Stop telling God what to do.”

Actually Einstein was not troubled by the indeterminism of quantum theory, but by the fact that he thought quantum theory was probably not the “ultimate” theory. He thought quantum theory was incomplete.

# John Bell



In 1964, John Bell showed that the question of whether or not quantum theory could be explained by a local hidden variable theory was an EXPERIMENTAL question (and thus a real scientific hypothesis!)

“the most profound discovery of science”

# A Game

Alice and Bob are locked away in prison cells and they cannot communicate.





A warden plays the following game.

1. He gives Alice and Bob a slip of paper. Written on each of these papers is the letter S or the letter T.

2. Alice are instructed that at a certain time, they will both be required to shout out either +1 or -1.



# A Game



Call Alice's output (+1,-1) if she received S, AS

Call Alice's output (+1,-1) if she received T, AT

Call Bob's output (+1,-1) if she received S, BS

Call Bob's output (+1,-1) if she received T, BT

| Alice's slip | Bob'slip | Product of their answers |
|---|---|---|
| S | S | ASBS |
| S | T | ASBT |
| T | S | ATBS |
| T | T | ATBT |

Note that the product of their answers depends only on what each party does locally.



# A Game



The Warden will let Alice and Bob free if they produce

| Alice's slip | Bob'slip | Winning Product |
|---|---|---|
| S | S | +1 |
| S | T | +1 |
| T | S | +1 |
| T | T | -1 |

Is it possible for Alice and Bob to always win this game?

No. ASBS= ASBT=+1 implies that BS=BT.

ATBS= -ATBT=+1 implies that BS=-BT.

But this implies BS=BT=0, contradiction!



# A Game



Suppose they play this game lots of times, each time they are given one of the sheets of paper with equal probability (i.e. the Warden gives each party an S or a T with 50% probability)

An indication of how well they are doing is to calculate the probability of winning:

Probability of winning= ¼(Pr(ASBS=+1|S,S)+ Pr(ASBT=+1|S,T)
+ Pr(ATBS=+1|T,S)+Pr(ATBT=-1|T,T))

How are they doing?



# A Game



Suppose Alice and Bob always play the same strategy.
This means assigning values to AS,AT,BS,and BT.

Probability of winning= ¼(Pr(ASBS=+1|S,S)+ Pr(ASBT=+1|S,T)
+ Pr(ATBS=+1|S,T)+Pr(ATBT=-1|S,T))

At most they can win ¾ of the time.
Why? There must always be a contradiction with the winning formula and so one of these probabilities is zero.

And they can achieve the case where one is zero and all three others are one by always having A and B output +1



# A Game

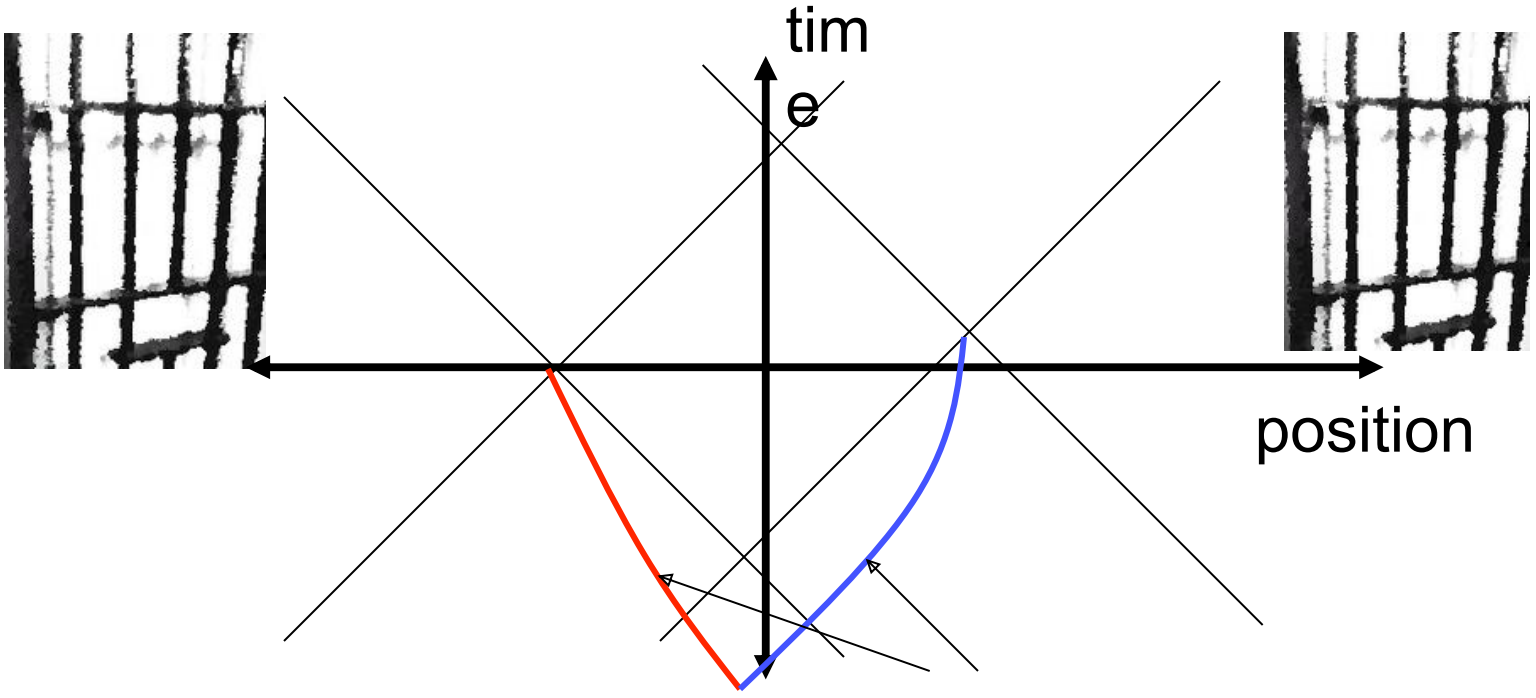


Suppose Alice and Bob are even allowed to share some random numbers they copied from each other before they were thrown in
jail.
What is their maximum probability of winning? Same as before.

Why? Fix the random numbers. Run the protocol.
Probability of winning will be less than ¾.
Average over the random numbers.
No help. Probability of winning is at most ¾.

Probability of winning is at most ¾

# The Connection




Local classical data
Local hidden variables

# The Connection

After the break: how quantum entanglement allows Alice and Bob a greater probability of freedom!

“The entanglement shall set you free!”



# A Game



Suppose Alice and Bob are even allowed to share an entangled quantum state which they made when they were plotting together before the were thrown in jail.

Suppose they each have one qubit of two qubits with wave function

$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

# A Basis

$$|a(\theta)\rangle = \cos(\theta)|0\rangle + \sin(\theta)|1\rangle$$

$$|b(\theta)\rangle = \sin(\theta)|0\rangle - \cos(\theta)|1\rangle$$

These states are orthogonal

$$\langle b(\theta)|a(\theta)\rangle = \cos(\theta)\sin(\theta) - \sin(\theta)\cos(\theta) = 0$$

And normal

$$\langle a(\theta)|a(\theta)\rangle = \cos(\theta)\cos(\theta) + \sin(\theta)\sin(\theta) = 1$$
$$\langle b(\theta)|b(\theta)\rangle = \sin(\theta)\sin(\theta) + (-\cos(\theta))(-\cos(\theta)) = 1$$

Measurement operators in this basis, labeling these outcomes + and -

$$M_+(\theta) = |a(\theta)\rangle\langle a(\theta)| \qquad M_-(\theta) = |b(\theta)\rangle\langle b(\theta)|$$

# A Measurement

$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

Suppose that Alice measures in the $|a(\theta)\rangle, |b(\theta)\rangle$ basis
and that Bob measures in the $|a(\phi)\rangle, |b(\phi)\rangle$ basis

Probabilities of outcomes:

$$Pr(+,+) = \langle v|M_+(\theta) \otimes M_+(\phi)|v\rangle$$

$$Pr(+,-) = \langle v|M_+(\theta) \otimes M_-(\phi)|v\rangle$$

$$Pr(-,+) = \langle v|M_-(\theta) \otimes M_+(\phi)|v\rangle$$

$$Pr(-,-) = \langle v|M_-(\theta) \otimes M_-(\phi)|v\rangle$$

# In Class Problem

# A Measurement

$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

Suppose that Alice measures in the $|a(\theta)\rangle, |b(\theta)\rangle$ basis
and that Bob measures in the $|a(\phi)\rangle, |b(\phi)\rangle$ basis

Probabilities of outcomes:

$$Pr(+,+) = \frac{1}{2}\cos^2(\theta - \phi)$$

$$Pr(+,-) = \frac{1}{2}\sin^2(\theta - \phi)$$

$$Pr(-,+) = \frac{1}{2}\sin^2(\theta - \phi)$$

$$Pr(-,-) = \frac{1}{2}\cos^2(\theta - \phi)$$



# A Game



They share $|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$

If Alice got a slip with S, she measures in the basis with $\theta = \theta_S$
If Alice got a slip with T, she measures in the basis with $\theta = \theta_T$

If Bob got a slip with S, she measures in the basis with $\phi = \phi_S$
If Bob got a slip with T, she measures in the basis with $\phi = \phi_T$

Both parties output their measurement (+ = +1, - =-1)

Choose angles which maximize the probability of winning

# A Measurement

$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

Suppose that Alice measures in the $|a(\theta)\rangle, |b(\theta)\rangle$ basis
and that Bob measures in the $|a(\phi)\rangle, |b(\phi)\rangle$ basis

Probabilities of outcomes:

$$Pr(+,+) = \frac{1}{2}\cos^2(\theta - \phi)$$

$$Pr(+,-) = \frac{1}{2}\sin^2(\theta - \phi)$$

$$Pr(-,+) = \frac{1}{2}\sin^2(\theta - \phi)$$

$$Pr(-,-) = \frac{1}{2}\cos^2(\theta - \phi)$$

$$Pr(+1) = \cos^2(\theta - \phi)$$

$$Pr(-1) = \sin^2(\theta - \phi)$$



# A Game



$$Pr(\text{win}) = \frac{1}{4}\left[\cos^2(\theta_S - \phi_S) + \cos^2(\theta_S - \phi_T) + \cos^2(\theta_T - \phi_S) + \sin^2(\theta_T - \phi_T)\right]$$

$$\theta_S = 0, \theta_T = \frac{\pi}{4} \qquad \phi_S = \frac{\pi}{8}, \phi_T = -\frac{\pi}{8}$$

$$Pr(\text{win}) = \frac{1}{4}\left[\cos^2\left(-\frac{\pi}{8}\right) + \cos^2\left(\frac{\pi}{8}\right) + \cos^2\left(\frac{\pi}{8}\right) + \sin^2\left(\frac{3\pi}{8}\right)\right]$$

$$Pr(\text{win}) = \frac{1}{2}\left(1 + \frac{1}{\sqrt{2}}\right) \approx 0.85355 > \frac{3}{4}$$



# A Game



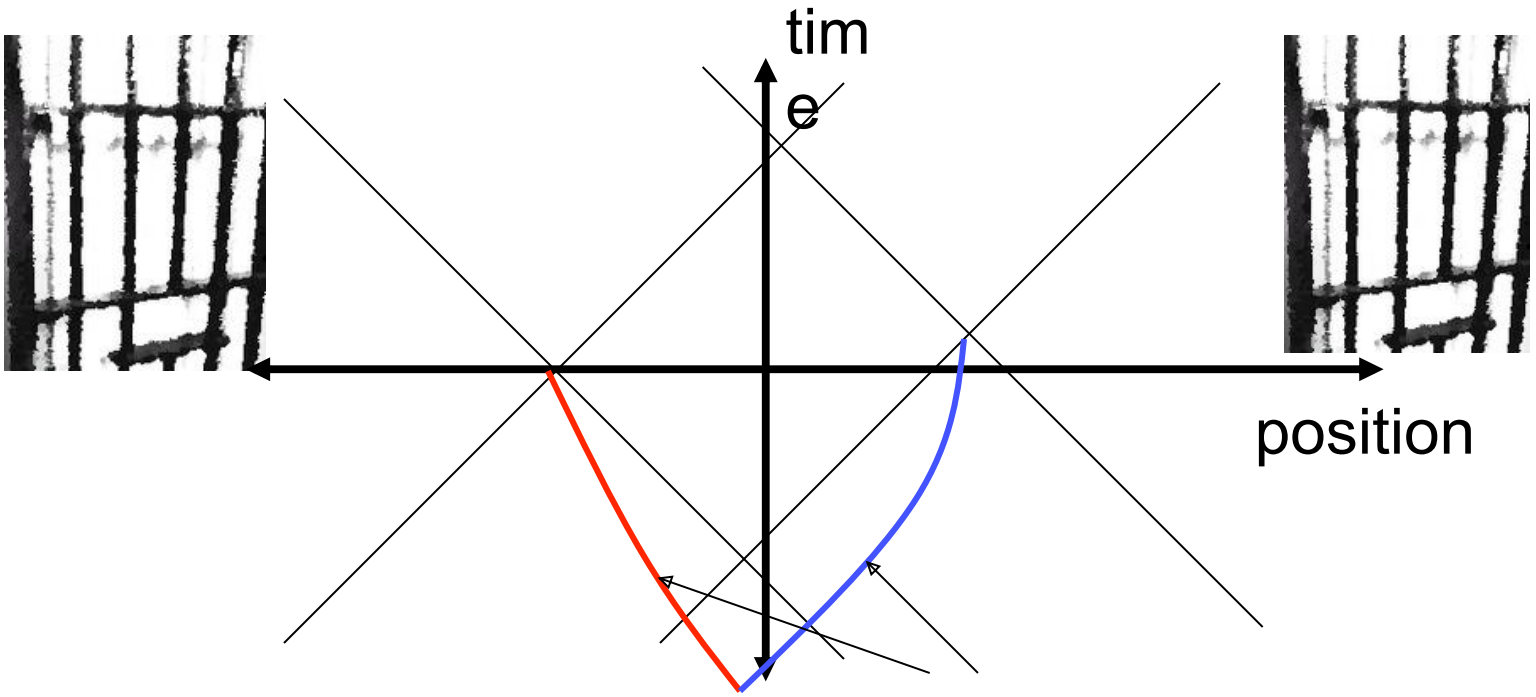
$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

By sharing this entangled state, which they cannot use to communicate with each other, but they can increase their chances of defeating this evil evil warden.

# Bell's Theorem

$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

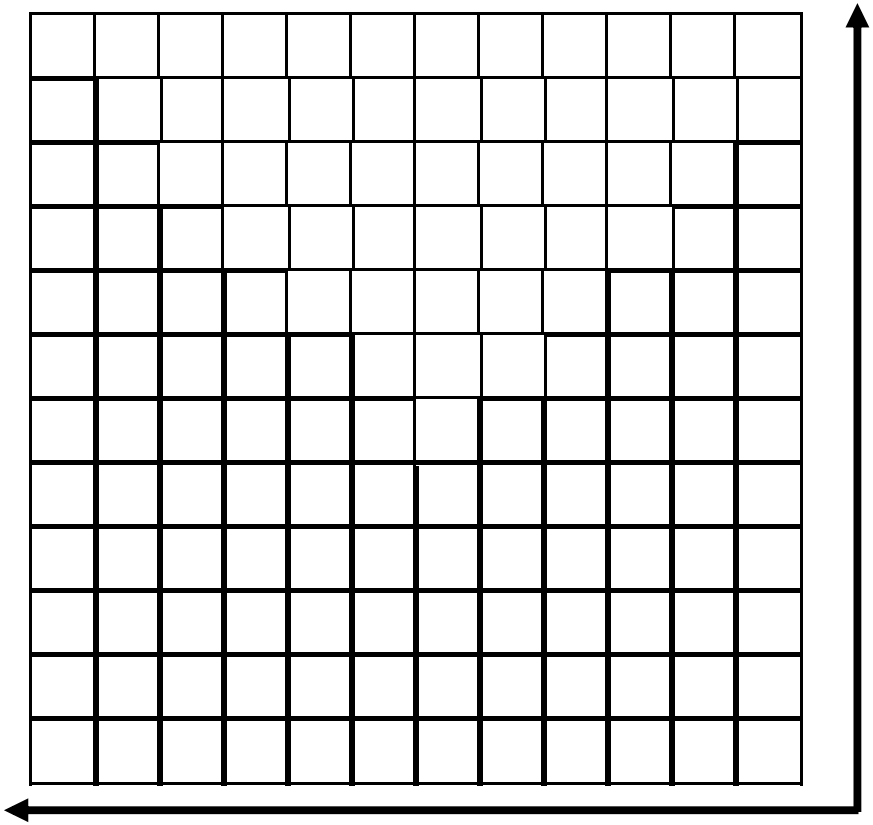
Bell's theorem tells us that we cannot simulate the statistics of quantum theory using a local hidden variable theory!




Local classical data
Local hidden variables

# Bell's Theorem

Why is this profound?



Local theories, with classical information, like cellular automata cannot reproduce the predictions of quantum theory

(Apologies to Stephen Wolfram)

# Bell's Theorem

Why is this profound?

Recall our model where we put bits into boxes to get correlation

Correlation is not profound.

But quantum correlation is different from experiments where we generate correlations locally!

This was truly the first result in quantum information:

quantum correlations play games better
than classical correlations

# Bell's Theorem

Why is this profound?

Bell's theorem is an experiment:

All local hidden variable theories satisfy:

$$Pr(\text{win}) \leq \frac{3}{4}$$

But quantum theory predicts

$$Pr(\text{win}) = \frac{1}{2}(1 + \frac{1}{\sqrt{2}}) > \frac{3}{4}$$

So go out and test it!!!! Famous tests of Aspect 1982.

**Experimental demonstration of quantum-correlations over more than 10 kilometers**

W. Tittel, J. Brendel, B. Gisin, T. Herzog, H. Zbinden, and N. Gisin
*University of Geneva, Group of Applied Physics, 20, Rue de l'Ecole de Médecine, CH-1211 Geneva 4, Switzerland*
*e-mail: wolfgang.tittel@physics.unige.ch*
(June 26, 2005)

Energy and time entangled photons at a wavelength of 1310 nm are produced by parametric down-conversion in a $KNbO_3$ crystal and are sent into all-fiber interferometers using a telecommunications fiber network. The two interferometers of this Franson-type test of the Bell-inequality are located 10.9 km apart from one another. Two-photon fringe visibilities of up to 81.6% are obtained. These strong nonlocal correlations support the nonlocal predictions of quantum mechanics and provide evidence that entanglement between photons can be maintained over long distances.

# Entanglement and Philosophy

Bell's theorem (along with another important result known as the Kochen-Specker theorem) is one of the strangest parts (some would say the only) of quantum theory.

Viewpoint 1: "Shut up and calculate", David Mermin

Quantum theory is consistent.
It provides no local realistic description, but that's just the way the world works.

Viewpoint 2: "Many worlds"

Those states in the superposition should be interpreted as different parallel universes.
Bell universes only split when communication of results between parties (split is always a local event)

There are hundreds more different viewpoints…..

# Entanglement and Philosophy

Until these different viewpoints yield different experimental results, we don't worry too much about them.

But: by studying the controversies, many quantum information scientists have developed a heightened awareness of what quantum theory does and does not say.

"I have this student, he's thinking about the foundations of quantum theory. He's doomed." –John McCarthy

"...contemporary physicists come in two varieties. Type 1 physicists are bothered by EPR and Bell's Theorem. Type 2 (the majority) are not, but one has to distinguish two subvarieties. Type 2a physicists explain why they are not bothered. Their explanations tend either to miss the point entirely (like Born's to Einstein) or to contain physical assertions that can be shown to be false. Type 2b are not bothered and refuse to explain why." – David Mermin

# Entanglement and Quantum Computation

We remarked before, that a general quantum state on n qubits

$$|\psi\rangle = \sum_{x=0}^{2^n-1} a_x |x\rangle$$

Required $2^n$ complex numbers to specify. Exponential in n!

But, suppose that this n qubit state is separable across all n qubits:

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle$$

Each single qubit wave function requires us to specify only two complex numbers, so we only need 2n complex numbers to specify this separable state!

# Entanglement and Quantum Computation

Suppose during an algorithm, the state always remains separable

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle$$

We can efficiently simulate this quantum computation on a classical computer:

One and two qubit gates are just 2x2 and 4x4 matrix multiplications.

So, we need some entanglement in a quantum algorithm in order to have a speedup of quantum over classical.

# Entanglement and Quantum Computation

A little more sophisticated [Jozsa and Linden]

p-blocked: If no p+1 subset of n qubits is entangled.

Such circuits can be efficiently simulated if p is poly(log n)

Basic idea: Entangled blocks of size p can be expressed as 2p complex numbers.

This is polynomial in n if p is poly(log n)

A “large amount” of entanglement is necessary for a quantum speedup. Interestingly it is not sufficient.

# Entanglement and Quantum Computation

A side benefit: physicists spend a lot of time trying to simulate quantum systems. This is often computationally hard.

Because we know that entanglement is important for this to be difficult, we can design our algorithms to keep track of as much entanglement as possible.

This has lead to nice deep insights into some algorithms physicists had previously invented, and is currently being exploited to design new faster algorithms!

Understanding entanglement is essential to understand the limits of computational quantum physics!

# Talk To Me



ALICE

a

Feq(a,b) =

1 if a=b
0 if a≠b



BOB

b

| 0 | 1 | 0 | 1 | ••• | 0 |
|---|---|---|---|---|---|

n bits

| 0 | 1 | 1 | 1 | ••• | 0 |
|---|---|---|---|---|---|

n bits

How much (many bits) do Alice and Bob need
to exchange to compute F(a,b)?

*With no error:* n bits

*With one-sided error ε:* O(log(n/ε)) bits

# Talk To Me Some More



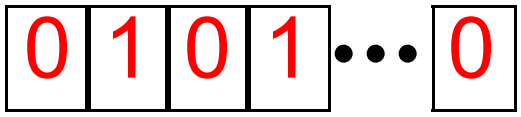
ALICE

a

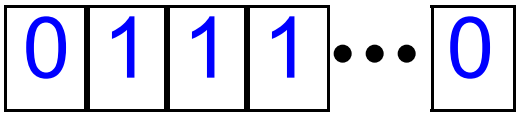


BOB

b




n bits




n bits

Ixor(a,b)=a1⊕a2 ⊕ …⊕an⊕b1⊕b2 ⊕ …⊕bn

How much (many bits) do Alice and Bob need to exchange to compute F(a,b)?

*With no error.*

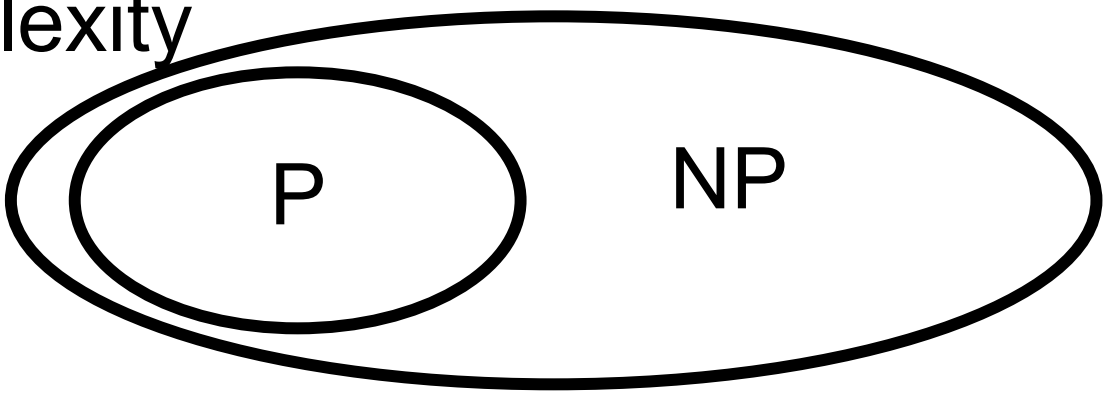
# Communication Complexity

How much communication between Alice and Bob is necessary for them to compute F(a,b) when Alice knows a and Bob knows b?

[Yao 79]

many rigorous lower and matching upper bounds

…in contrast to computational complexity



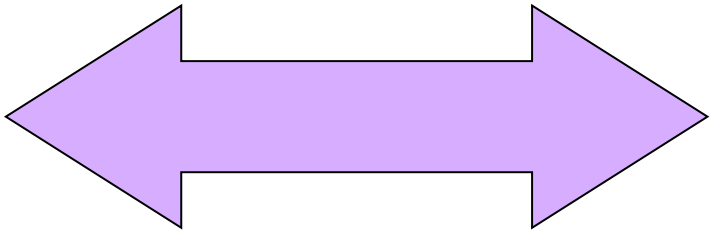

N=1
?
P=0?

# Make It Quantum



ALICE

a



communicate qubits

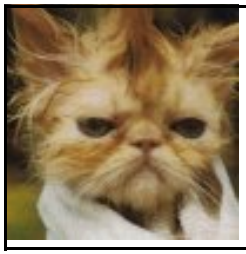
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$



BOB

b

How does classical communication complexity (bits) compare to quantum communication complexity (qubits)?



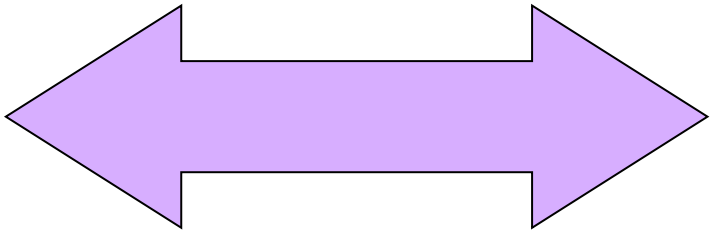
**Holevo's Theoerem:** sending n classical bits requires n quantum bits.

# Holevo's Theorem



ALICE

a



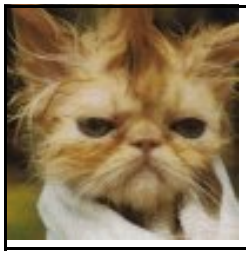
communicate qubits

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$



BOB

b



**Holevo's Theoerem:** sending n classical bits requires n quantum bits.

One is often tempted to think about a qubit as storing an analog number, the amplitudes. But this is totally wrong. This would be like saying that you are storing a an infinite amount of information in a classical bit to which you have probabilities for the state of the bit

# Not So Fast



Ran Raz
(1999)

There exists functions F(a,b) which require Ω(m1/4/log(m)) bits of communication to solve
but can be solved using only O(log(m)) qubits.
[m is the length of a and b]



Exponential benefit in quantum over classical.

Exact model 0-error:
Buhrman, Cleve, Wigderson (1998)
1-way communication exponential separation:
Bar-Yossef, Jayram, Kerenidis (2004)

# Quantum Teleportation



ALICE

a

1 qubits

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$



BOB

b

$\cong$



ALICE

a

2 bits

$|\Psi_-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

entangled quantum state



BOB

b

Quantum Teleportation Protocol

# Entanglement as a Resource

If parties share entangled states, they can outperform classical communication complexity protocols.

But there are all sorts of different “kinds” of entanglement.

For example, for two qubits:

$$|\Psi\rangle = \cos(\theta)|00\rangle + \sin(\theta)|11\rangle$$

When $\theta = \frac{\pi}{4}$

,

$$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

“maximally entangled”
“1 ebit”

If we use non-maximally entangled qubits for teleportation, for example, it becomes less efficient.

# Entanglement Distillation

Suppose I give you many copies of a non-maximally entangled qubits

$$|\Psi\rangle = \cos(\theta)|00\rangle + \sin(\theta)|11\rangle$$

Is it possible to convert $|\Psi\rangle^{\otimes n}$ into m<n maximally entangled qubits?

$$|\Psi\rangle^{\otimes n} \rightarrow |\Phi_+\rangle^{\otimes m}$$

Entanglement distillation! Rate is

$$\frac{m}{n} = S(\rho) = -\cos^2(\theta)\ln\cos^2(\theta) - \sin^2(\theta)\ln\sin^2(\theta)$$

von Neumann entropy

# Entanglement Dilution

What about the reverse? n>m

$$|\Phi_+\rangle^{\otimes m} \rightarrow |\psi\rangle^{\otimes n}$$

Entanglement diluation

$$\frac{m}{n} = S(\rho) = -\cos^2(\theta)\ln\cos^2(\theta) - \sin^2(\theta)\ln\sin^2(\theta)$$

von Neumann entropy

# Fungible

(pure state) bipartite entanglement is a fungible resource which can be quantified in the quantity of ebits.

Fungible, adj. 1. being of such a nature that one part or quantity may be replaced by another equal part or quantity in the satisfaction of an obligation

# Entanglement

Bell's theorem says quantum theory is not a local realistic hidden variable theory

Entanglement can be used to speed up communication complexity problems over classical problems

Entanglement is necessary (but not sufficient) for speedups in quantum algorithms

Entanglement is a fungible resource measured in ebits